NAME: _____

Lab 3 — Standing Waves on a String

SYNOPSIS

In Note 1 you drew the modes of a string fixed at both ends and wrote down that each one adds half a wave, so that $f_2 = 2f_1$, $f_3 = 3f_1$, and so on. Those were drawings. In this lab you make them appear on a real string and read their frequencies off a dial. You will find the fundamental, climb the harmonics one at a time, and plot frequency against harmonic number — and the slope of that plot is a number you have already predicted, which is what makes the plot worth making. Then you use one harmonic to get a wavelength, combine it with its frequency, and obtain the speed of a wave on the string.

Performance Expectations: Set up a driven string under measured tension and locate at least the first five resonances by adjusting the driving frequency. Record harmonic number and frequency as a data table, plot frequency against harmonic number, fit a trendline, and interpret its slope. Determine a wavelength from the geometry of a named harmonic and compute the wave speed from $v = \lambda f$. Predict and then test how the tension changes the wave speed and the resonant frequencies. Distinguish what you measured from what you inferred, and cite your own data when you do.

Equipment: Pasco economy wave driver · Pasco sine wave generator · about 1.75 m of Mason's twine · pulley and bracket · small lab stand · mass set (20 g, 50 g, 100 g, 200 g) · metre stick or tape measure · a laptop with Excel for the plot.

Setting up the apparatus

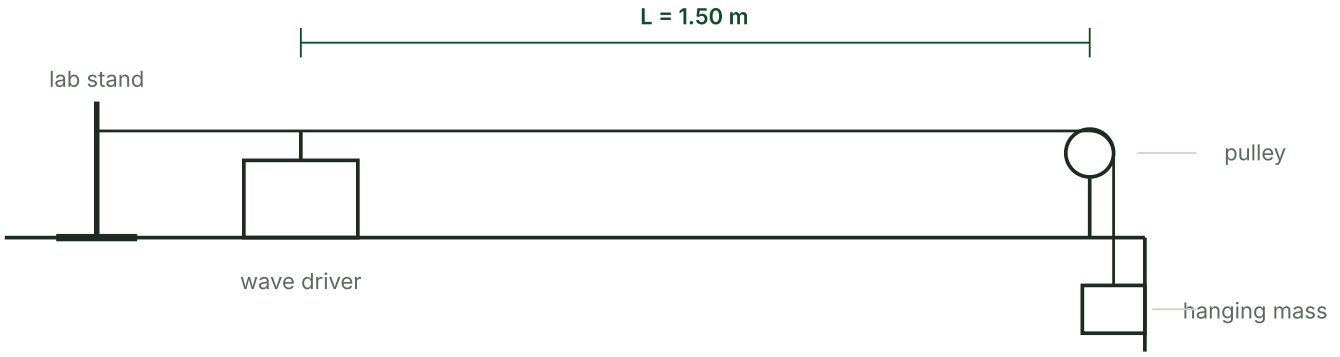


Figure 1. The apparatus. The vibrating length L is measured from the driver post to the centre of the pulley, **not** from the lab stand — the segment behind the driver is not part of the standing wave.

1. Clamp the pulley and bracket firmly to the edge of the table. Place the wave driver about 1.5 m back from it along the table.

2. Loop the string through the economy wave driver and tie the far end off to the small lab stand. Run the other end over the pulley and hang **200 g** from it, making sure the mass hangs freely and is not resting on anything.

3. Slide the wave driver so that the driver post and the centre of the pulley are **1.50 m** apart. Adjust the pulley height until the string is horizontal.

4. Measure and record L and the hanging mass. Then make your own sketch of the apparatus below, labelling the equipment and marking every distance and force you can identify. You will need this sketch in your report.

YOUR SKETCH OF THE APPARATUS

Vibrating length L (m)	Hanging mass (g)	Tension = mg (N)	String
			Mason's twine

BEFORE YOU TURN ANYTHING ON

Turn the amplitude down to about ¼ of maximum and leave it there. A large amplitude makes the fundamental easy to see and every harmonic above it hard to find, because the string whips instead of settling. If the mass is swinging, stop it with your hand before taking a reading.

Part 1 — Find the fundamental

1. Set the frequency to about 10 Hz. Raise it slowly, watching the string.

2. At some point the string swings up and down as **one single loop**, with the largest motion at the centre and no motion at the ends. Fine-tune with the 1.0 Hz and 0.1 Hz knobs until the amplitude is as large as you can make it. That frequency is the fundamental, f_1.

   f_1 = _____ Hz

3. **Before you go on**, write down what you expect f_2 and f_3 to be, from Note 1. You are about to measure them.

   predicted f_2 = _____ Hz **predicted f_3 =** _____ Hz

IF YOU CANNOT FIND IT

It is often easier to find the **two-loop** pattern first, because it is unmistakable, and then halve the frequency to land on the fundamental. If the whole string is just shaking without a clean shape anywhere, your amplitude is too high or the mass is swinging.

Part 2 — Climb the harmonics

Keep raising the frequency. Each time the string settles into a pattern with one more loop than before, fine-tune for maximum amplitude and record it. You should reach at least the fifth harmonic without difficulty; after that the pattern gets harder to see, but take as many as you can.

Harmonic number n	Number of loops	Measured frequency f (Hz)	f / n (Hz)
1			
2			
3			
4			
5			
6			
7			

That last column is a check you can run while you are still at the bench. If the values down it are not all about the same, one of your rows is mislabelled — most likely you skipped a harmonic without noticing.

Part 3 — Analysis

A. Frequency against harmonic number

1. Enter your table into Excel and plot **frequency on the vertical axis against harmonic number on the horizontal axis**. Label both axes with quantity and unit, and give the plot a title.

2. Add a linear trendline and display its equation. Record the slope.

   slope = _____ Hz per harmonic

3. Compare the slope with one of the numbers already in your data table. Which one is it, and why should it be that one?

B. Wavelength and wave speed

Use the **second** harmonic, where the string carries two loops across the length L.

1. Each loop is half a wavelength. Write down the wavelength of the second harmonic in terms of L, then put your measured L in.

   λ_2 = _____ m

2. Combine it with the frequency you measured for that harmonic to get the speed of the wave on the string.

   v = _____ m/s

3. Repeat for one other harmonic of your choice. Do you get the same v? Should you?

   n = _____ λ_n = _____ m v = _____ m/s

$$v = \lambda f$$

The same relation you used in Lab 2 to turn a measured wavelength into a frequency — run the other way here. **Include a sample calculation** for every type of calculation in your report: starting equation, algebra, result, units at every step.

Part 4 — Change the tension

Everything above was taken at one tension. Now change it.

1. **Predict first, in writing.** You are going to halve the hanging mass. State what you expect to happen to the speed of the wave on the string, and what you expect to happen to the fundamental frequency.

   v will: _____

   f_1 will: _____

2. Replace the 200 g with 100 g. Find the fundamental again, and the second harmonic.

3. Compute the new wave speed from the second harmonic, exactly as in Part 3B. The length has not changed, so λ_2 has not changed either.

4. Fill in the comparison. Was your prediction right, in direction and in rough size?

Hanging mass	Tension (N)	f_1 (Hz)	f_2 (Hz)	λ_2 (m)	v (m/s)
200 g					
100 g					

Questions for Discussion

Choose three of the following and answer those three fully, working them into your discussion section. Say which three you chose. Do not simply copy the question and answer it in a line — and do not limit yourself to these topics if you have other insights.

1. What happened to the wavelength as the frequency increased? Express the relationship between the two as a proportionality, and name what was held constant to make it true.

2. Why is the slope of your Part 3A graph the number it is? Answer in terms of the shapes on the string, not in terms of the trendline.

3. The ends of the string are the one place where nothing moves. Explain why that single constraint is what forces the frequencies to come in a whole-number sequence instead of arbitrary values.

4. From your Part 4 data: which of v, λ and f changed when you changed the tension, and which did not? Use that to say what physical property of the system the tension actually sets.

5. The frequencies you found were the ones where the string responded with a large amplitude; at every other setting it barely moved. Connect that to the resonance argument you made about the Tacoma Narrows bridge in Note 1.

6. A guitar string is tuned by turning a peg, and fretted by shortening it. Using your own two experiments, say which of those two changes v and which changes λ, and explain how each ends up changing the pitch.

> **LAB REPORT**
>
> Follow the standard format in the **Writing Lab Reports** module. The report template for this lab gives fewer opening sentences than Lab 2 did — that is deliberate, and the template says so on its first page. Your abstract must state your measured wave speed and the tension it was measured at.

Copyright © 2026 by Dr. Byungmoon Cho · FLCC Physics · PHY 105